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# Predicting species richness and diversity

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Questions so far?

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## Diversity as a community property

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**If we model the community, diversity follows.**

## Why model-based diversity?

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Without a generative model:

- ▶ No uncertainty quantification: no standard errors, no confidence intervals
- ▶ No ability to account for sampling: effort, detection, protocol
- ▶ No ability to account for data properties: overdispersion, zero-inflation
- ▶ No prediction for new sites, times, or conditions

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A community model addresses all of these: diversity follows from the generative model and inherits its inferential properties.



## What is species richness?

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It is by far the most common biodiversity metric: simple to calculate, intuitive to interpret, and widely used in conservation and ecology.

But it is a **derived quantity** from multivariate community data. Analyzing it directly means throwing away most of the data.

## The typical approach

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Species richness is most commonly analyzed with Poisson or negative-binomial regression:

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Species richness is most commonly analyzed with Poisson or negative-binomial regression:

$$\log\{\mathbb{E}(SR_i)\} = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} \quad (2)$$

This is simple and interpretable. But it rests on four assumptions that are routinely violated in practice:

1. Species are **independent** of each other
2. All species respond **identically** to the environment
3. The species pool is **infinite** (unbounded distribution)
4. The Poisson or NB distribution is **appropriate**

## Assumption 1: species independence

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If species co-occurrence is positive, species richness has **less** variance than a Poisson. If it is negative, species richness has **more** variance.

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Species are not independent; they share environmental responses and co-occur non-randomly.

If species co-occurrence is positive, species richness has **less** variance than a Poisson. If it is negative, species richness has **more** variance.

A Poisson or NB regression cannot accommodate this: it will over- or underestimate uncertainty.



## Assumption 2: homogeneity of niches

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A Poisson or NB distribution for species richness is only justified if all species respond the same way to the environment, i.e., have the same niche.

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## Assumption 2: homogeneity of niches

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A Poisson or NB distribution for species richness is only justified if all species respond the same way to the environment, i.e., have the same niche.

This is ecologically unrealistic: species have different distributions, different ranges, different optimums.

The consequence: estimated coefficients from a Poisson richness regression are a **weighted average** over species, with weights depending on the distribution of species' niches, usually not the average ecologists want.

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**This is ecologically impossible.** You cannot observe more species than those you have observed.

Negative-binomial adds overdispersion, making the problem worse.

Species richness is inherently bounded by the number of species in the observed pool.

## The correct distribution: Poisson-Binomial

---

Under species independence and Bernoulli responses, species richness follows a **Poisson-Binomial distribution**:

$$SR_i \sim \mathcal{PB}(\pi_{i1}, \pi_{i2}, \dots, \pi_{im}) \quad (3)$$

with

$$\mathbb{E}(SR_i) = \sum_{j=1}^m \pi_{ij}, \quad \text{var}(SR_i) = \sum_{j=1}^m \pi_{ij}(1 - \pi_{ij}) \quad (4)$$

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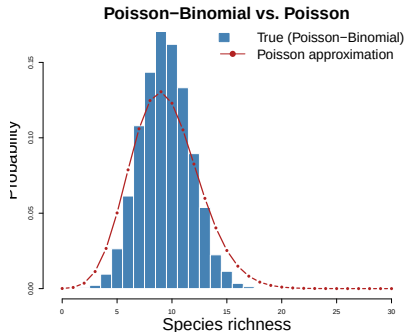
with

$$\mathbb{E}(SR_i) = \sum_{j=1}^m \pi_{ij}, \quad \text{var}(SR_i) = \sum_{j=1}^m \pi_{ij}(1 - \pi_{ij}) \quad (4)$$

- ▶ Mean richness = **sum of species' occurrence probabilities**
- ▶ Variance is bounded (cannot exceed  $m/4$ )
- ▶ This is the distribution that actually generates species richness
- ▶ The Binomial is a special case (all  $\pi_{ij}$  equal)



# Why univariate analysis fails



The Poisson distribution:

- ▶ Puts probability mass on impossible values ( $SR > m$ )
- ▶ Has the wrong shape (underdispersed here)
- ▶ Assumes the wrong mean-variance relationship

The mismatch worsens when species niches are heterogeneous.

## Species richness as an emergent property

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We do not model species richness directly; instead, **species richness emerges** from each species' response to the environment:

$$\mathbb{E}(SR_i) = \sum_{j=1}^m \mathbb{E}\{I(y_{ij} > 0)\} = \sum_{j=1}^m p(y_{ij} > 0) \quad (5)$$

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- ▶ No separate richness model needed
- ▶ Works for any data type: binary, counts, cover, biomass, ordinal
- ▶ Uncertainty propagates naturally from species-level predictions
- ▶ Assumptions are explicit and testable
- ▶ Non-linear richness gradients emerge even from linear species responses
- ▶ Inherits whatever structure the species model has: traits, phylogeny, spatial or hierarchical effects

## Works for any data type

---

For any distribution with zero support, the probability of presence can be derived:

| Data type | Distribution      | $p(y_{ij} > 0)$                         |
|-----------|-------------------|-----------------------------------------|
| Binary    | Bernoulli         | $\pi_{ij}$                              |
| Count     | Poisson           | $1 - e^{-\mu_{ij}}$                     |
| Count     | Negative-binomial | $1 - (1 + \mu_{ij}/k)^{-k}$             |
| Biomass   | Tweedie           | $1 - \text{Tw}(0; \mu_{ij}, \phi_j, p)$ |
| Cover     | Ordered beta      | $1 - \rho_{ij}^0$                       |

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The only requirement: the distribution must assign probability mass at zero. Gamma and beta distributions do not.

## Implementation in gllvm

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The `predictSR()` function computes:

- ▶ Expected species richness  $\hat{S}_i = \sum_j \hat{p}_{ij}$
- ▶ Full Poisson-Binomial distribution at each site
- ▶ Prediction intervals; optionally for new predictor values
- ▶ Per-species AUC via `goodnessOfFit()`: well-calibrated  $\hat{p}_{ij}$  give well-calibrated  $\hat{S}_i$

```
sr_pred <- predictSR(model, newX = newdata,  
                     newLV = matrix(0, ncol = 2, nrow = nrow(  
sr_pred$expected      # expected species richness  
residuals(sr_pred, model) # Dunn-Smyth residuals  
goodnessOfFit(Y, object = model, measure = "AUC") # AUC per s
```

## Example: alpine plants in Switzerland

---

- ▶ Data by D'Amen et al. (2018)
- ▶ Occurrence of 175 species at 840 plots
- ▶ Binary data (presence-absence)
- ▶ Sampled on an elevation gradient in Switzerland

```
dim(Y)
```

```
## [1] 840 175
```

## Example: fit a JSMD

---

```
## gllvm
##      12
## attr(,"autopar")
## [1] TRUE
```

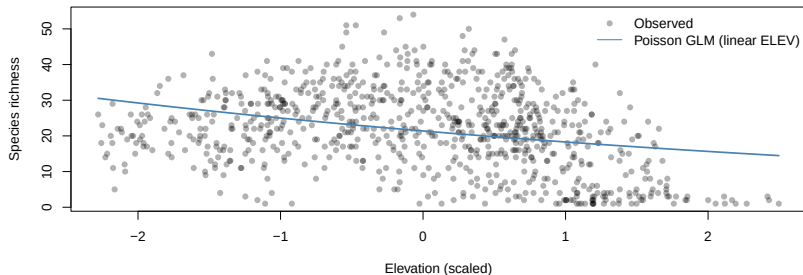
```
model <- gllvm(y = Y[,c(59,(1:ncol(Y))[-59])], X = X_scaled,
               formula = ~ELEV,
               num.lv = 2,
               family = "binomial",
               sd.errors = FALSE)
```

We include elevation as a fixed covariate and two latent variables to capture residual species associations.



## Example: a Poisson GLM for species richness

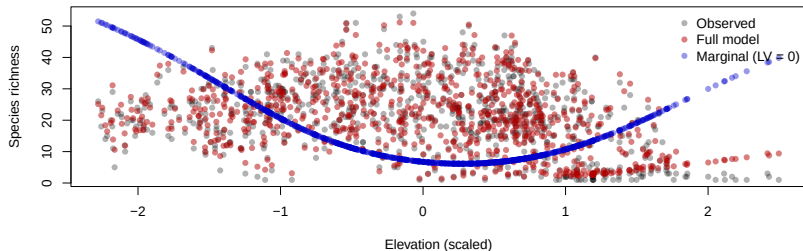
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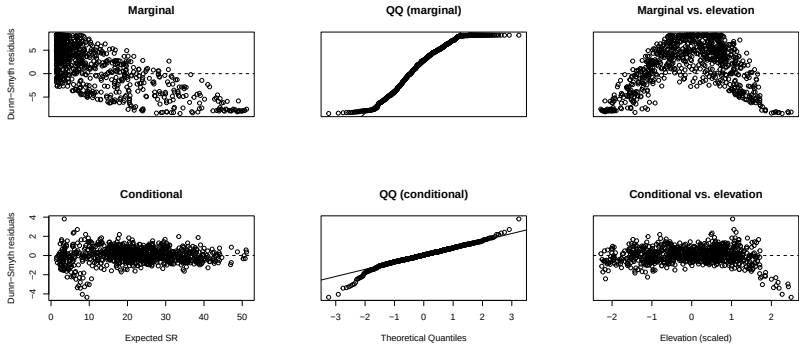
A Poisson GLM with linear elevation predicts a monotone richness curve. Does the multispecies model agree?

## Example: LV-elevation confounding

```
obs_sr <- rowSums(Y)
sr_full <- predictSR(model, se.fit = FALSE)
sr_marg <- predictSR(model, se.fit = FALSE, newLV = matrix(0, ncol = 2, nr
```



## Example: SR residuals, linear elevation term



Marginal residuals ( $LV = 0$ ) reveal the arch the linear fixed effect misses; conditional residuals are flat because the latent variables absorb the curvature.

## Example: adding a quadratic term

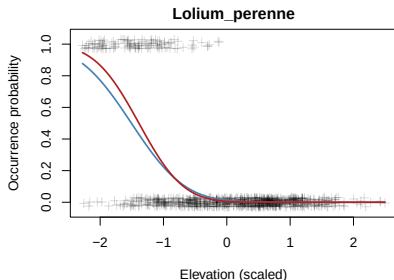
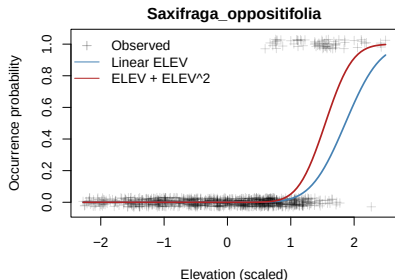
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```
model_lin <- model
model <- gllvm(y = Y[,c(59,(1:ncol(Y))[-59])], X = X_scaled,
              formula = ~ELEV + I(ELEV^2),
              num.lv = 2,
              family = "binomial",
              sd.errors = TRUE)
```

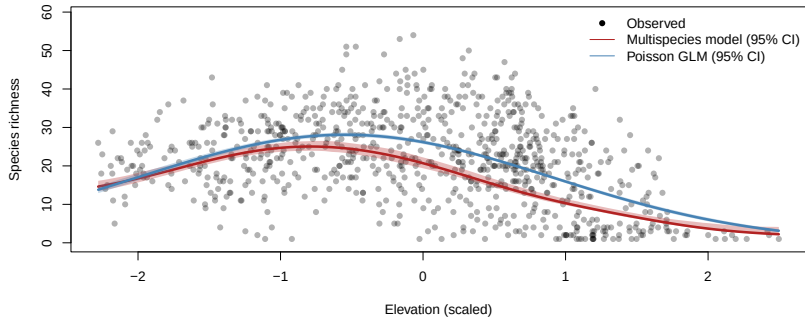
Species richness peaks at intermediate elevation. Without  $ELEV^2$  in the fixed effects, the latent variables absorb the curvature and new-site predictions fail. Adding the quadratic keeps the nonlinearity in the fixed effects where it belongs.

## Example: why `plot(model)` does not reveal this

- ▶ `plot(model)` uses **conditional** fitted values: latent variables are estimated per site and adapt to each model
- ▶ A missing quadratic term is absorbed by the latent variables; conditional residuals look clean for both models
- ▶ Marginal predictions ( $LV = 0$ ) reveal the discrepancy

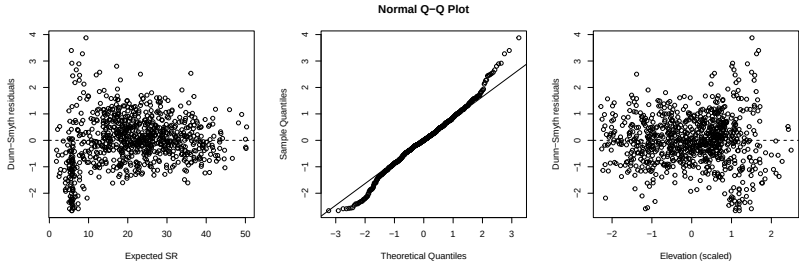


## Example: comparing predictions



**Question:** Both models include a quadratic elevation term. Why do their predictions differ?

## Example: residual diagnostics



Poisson-Binomial quantile residuals: no systematic elevation trend remains after adding the quadratic term.

## From richness to diversity

---

Species richness is one dimension of diversity; the model gives us a tractable distribution for it.



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These all follow from the **variance structure of  $\eta_{ij}$** ; like  $R^2$  in mixed models (Nakagawa & Schielzeth 2013), we work on the link scale rather than the response scale.

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But diversity has many facets: how dominant is the community at a site? How similar are nearby sites? How variable is composition across the region?

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Unlike species richness, `gllvm` does not (yet) have a dedicated function for these: what follows is computed directly from the fitted model's variance components, not a canned implementation. Still worth working through, since the ingredients are already sitting in every model you fit.

## Alpha, beta, and gamma diversity

---

- ▶ **Alpha:** local diversity at a single site; how uneven is the community here?
- ▶ **Gamma:** regional diversity; how much variation is there across the whole community?
- ▶ **Beta:** turnover between sites; how different are two communities?

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The following slides present calculations for a specific model.  
Results are model-dependent.

## Dominance

---

For  $\eta_{ij} = \beta_{0j} + \mathbf{x}_i^\top \boldsymbol{\beta}_j$  with  $\beta_{0j} \sim \mathcal{N}(0, \sigma_{\beta_0}^2)$  and  $\boldsymbol{\beta}_j \sim \mathcal{N}(\mathbf{0}, \Sigma_\beta)$ :

$$D_i = \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_i \quad (6)$$

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$$D_i = \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_i \quad (6)$$

- ▶ Large  $D_i$ : strong environmental filtering; a few well-adapted species dominate
- ▶ Small  $D_i$ : weak environmental sorting; more even composition



## Gamma diversity

---

Gamma diversity is the expected within-site variation over sites:

$$\gamma = \sigma_{\beta_0}^2 + \mathbb{E}_i(D_i) \quad (7)$$

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Regional diversity decomposes into a baseline  $\sigma_{\beta_0}^2$  (species differ even at the environmental centroid) and average dominance  $\mathbb{E}_i(D_i)$  (the contribution of environmental sorting across sites).

## Pairwise similarity (beta diversity)

---

Pairwise similarity between sites  $i$  and  $k$  is the covariance of the linear predictor across species:

$$\text{cov}_j(\eta_{ij}, \eta_{kj}) = \sigma_{\beta_0}^2 + \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k \quad (8)$$

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- ▶  $\sigma_{\beta_0}^2$ : baseline similarity shared by all site pairs
- ▶  $\mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k$ : environmental similarity; large when sites share the same environment

## Turnover

---

Turnover between sites  $i$  and  $k$  follows from the covariance structure:

$$\begin{aligned}\text{var}_j(\eta_{ij} - \eta_{kj}) &= \text{var}_j(\eta_{ij}) + \text{var}_j(\eta_{kj}) - 2 \text{cov}_j(\eta_{ij}, \eta_{kj}) \\ &= \sigma_{\beta_0}^2 + D_i + \sigma_{\beta_0}^2 + D_k - 2 \left( \sigma_{\beta_0}^2 + \mathbf{x}_i^\top \Sigma_\beta \mathbf{x}_k \right) \\ &= (\mathbf{x}_i - \mathbf{x}_k)^\top \Sigma_\beta (\mathbf{x}_i - \mathbf{x}_k)\end{aligned}\tag{9}$$

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The baseline  $\sigma_{\beta_0}^2$  cancels. Turnover has the same quadratic form as  $D_i$ , evaluated at the difference  $\mathbf{x}_i - \mathbf{x}_k$ .

## Uniqueness

---

The average pairwise similarity of site  $i$  with all other sites is:

$$\mathbb{E}_k \{ \text{cov}_j(\eta_{ij}, \eta_{kj}) \} = \sigma_{\beta_0}^2 + \mathbf{x}_i^\top \Sigma_\beta \bar{\mathbf{x}} \quad (10)$$

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**Uniqueness** is the average dissimilarity of site  $i$  from all other sites:

$$\text{uniqueness}_i = \mathbb{E}_k \{ \text{var}_j(\eta_{ij} - \eta_{kj}) \} = \sigma_{\beta_0}^2 + D_i + \gamma - 2 \left( \sigma_{\beta_0}^2 + \mathbf{x}_i^\top \Sigma_\beta \bar{\mathbf{x}} \right) \quad (11)$$



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Sites at environmental extremes have high uniqueness: their communities differ most from the regional average, potential refugia, range edges, endemic communities.

## Example: Swiss birds

---

Swiss breeding bird atlas: 56 species at 1000+ sites, with temperature (bio\_1), precipitation (bio\_12), and slope.

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Fit a JSMDM with random species-specific slopes, no LVs:

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model_birds <- gllvm(Y, X, formula = ~(bio_1 + bio_12 + slp | 1),  
                     family = "binomial", num.lv = 0, beta0com = TRUE)
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```

$\hat{\Sigma}_{\beta}$  is then the estimated covariance of random slopes. All diversity quantities follow from it.

## Predicting dominance

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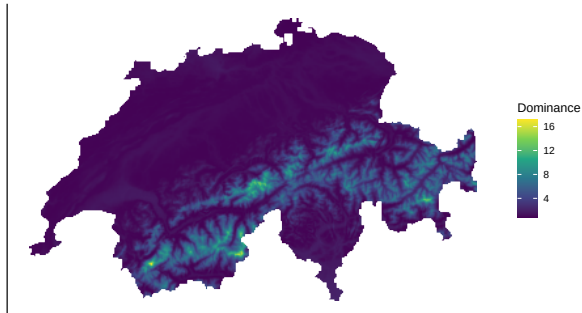
```
terra::values(alpha) <- rowSums(x * (x %*% model_birds$params$sigmaB))
```

`model_birds$params$sigmaB` is  $\hat{\Sigma}_{\beta}$ : the estimated covariance matrix of random slopes. This evaluates  $D_i = \mathbf{x}_i^{\top} \hat{\Sigma}_{\beta} \mathbf{x}_i$  at each grid cell: the environmental component of alpha diversity (dominance) due to temperature, precipitation, and slope.

# Dominance of Swiss birds

---

Bird dominance (due to temperature, precipitation and slope)



## Extending to LVs

---

Adding  $\mathbf{u}_i^\top$  to the linear predictor introduces residual species associations not captured by environment. Each diversity quantity gains an extra term:

- ▶ **Within-site variation:**  $\sigma_{\beta_0}^2 + D_i + \mathbf{u}_i^\top \frac{1}{m} \Lambda^\top \Lambda \mathbf{u}_i$
- ▶ **Turnover:**  $\Delta \mathbf{x}^\top \Sigma_\beta \Delta \mathbf{x} + \Delta \mathbf{u}^\top \frac{1}{m} \Lambda^\top \Lambda \Delta \mathbf{u}$
- ▶ **Avg. pairwise similarity:**  $\sigma_{\beta_0}^2 + \mathbf{x}_i^\top \Sigma_\beta \bar{\mathbf{x}}$ , unchanged (LV scores are centered,  $\bar{\mathbf{u}} \approx \mathbf{0}$ )

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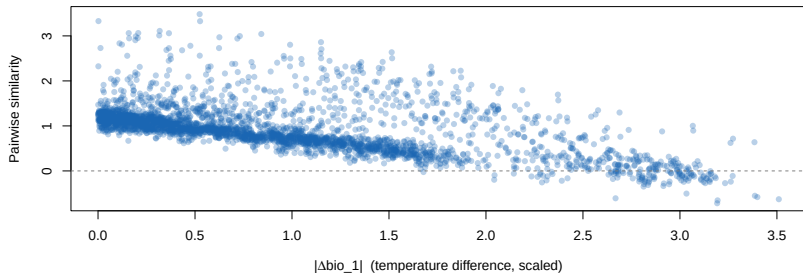
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LVs primarily inflate uniqueness (through alpha) and add residual turnover. Gamma and uniqueness extend analogously.



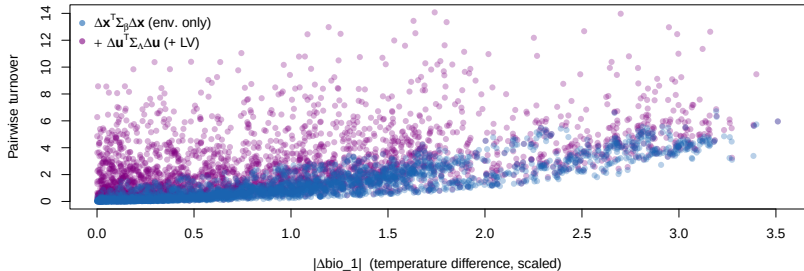
## Example: pairwise similarity

---



Pairwise similarity decreases with temperature difference. Pairs at opposite ends of the gradient can be negative: their species compositions are anti-correlated, not merely dissimilar.

## Example: pairwise turnover

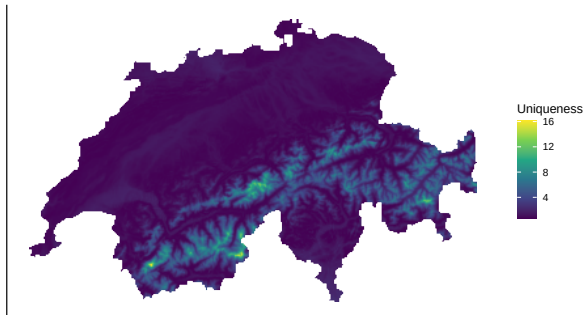


Environmental turnover increases with temperature difference; the LV component adds residual community dissimilarity not explained by the measured predictors. Averaging pairwise turnover over all partners  $k$  gives uniqueness: the per-site expected departure from the rest of the landscape.

## Example: uniqueness

---

Bird uniqueness (environmental distance from regional average)



Sites at environmental extremes are the most unique: their communities differ most from the regional average. These are the sites that drive beta diversity.

## Summary

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- ▶ Model the community; diversity follows
- ▶ Univariate species richness analyses make strong and routinely violated assumptions
- ▶ Variance components connect to alpha, beta, and gamma diversity;  $D_i$  captures environmental dominance
- ▶ Uniqueness quantifies how much a site's community differs from the regional average

## Summary

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- ▶ Model the community; diversity follows
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- ▶ Variance components connect to alpha, beta, and gamma diversity;  $D_i$  captures environmental dominance
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**Modelling species richness well requires modelling species well first.**